Ehsan Sharafian

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PROFESSIONAL SUMMARY

Mechanical Engineering PhD candidate developing intelligent wearable systems for human movement analysis and digital health. Experienced in building end-to-end solutions that integrate embedded hardware, IMU-based sensing, smartphone applications, biomechanics, machine learning, and haptic feedback for real-time gait analysis and personalized feedback. Proficient in Python, MATLAB, Android Studio/Java, PyTorch, and wearable system development, with six peer-reviewed journal publications and a strong focus on translating research into deployable medical and digital health technologies.

TECHNICAL SKILLS

Programming & Data Analysis	Python, Java, C/C++, NumPy, Pandas, scikit-learn, PyTorch
Mobile & Software	MATLAB, Simulink, Android Studio, Git
Wearable & Embedded Systems	IMU sensors, Arduino IDE, SparkFun modules, vibrotactile motors, PCB modules, Altium Designer
Biomechanics & Signal Processing	Gait analysis, human activity recognition, sensor fusion, time-series analysis, feature extraction, OpenSim
Robotics	Surgical robotics, parallel manipulators, inverse kinematics, dynamics, control systems, trajectory planning
CAD	SolidWorks

RESEARCH & ENGINEERING EXPERIENCE

Graduate Research Assistant | Biorobotics and Biomechanics Lab, University of Maine — Sep 2023 – Present
Orono, ME

- Designed and validated a real-time, smartphone-deployed gait assessment system using a single foot-mounted IMU to reconstruct foot-height trajectories and classify level walking, stair ascent/descent, and ramp ascent/descent with 96.08% accuracy, less than 1.1 cm cumulative height error, and 112 ms latency.
- Engineered a lightweight human activity recognition system using two IMUs and machine learning techniques to detect normal locomotion with 99.1% accuracy and classify atypical stair-negotiation strategies in older adults with 93.1% accuracy and 53 ms latency.
- Developed a home-based haptic feedback gait-training system integrating IMUs, vibrotactile motors, custom Android applications, Arduino/SparkFun modules, and PCB circuitry to provide real-time thigh-extension feedback for adults aged 65+, increasing stride length and walking speed by up to 30% in a single-session study with minimal cognitive demand.
- Validated wearable systems with human participants, including 40 adults aged 65+ and 2 stroke patients in a haptic feedback gait study, 20 young adults in a foot-clearance study, and 10 young adults plus 15 older adults in activity recognition experiments.
- Implemented real-time algorithms and data analysis pipelines in Android Studio/Java, Python, and MATLAB, combining signal processing, sensor fusion, biomechanical modeling, and machine learning for mobile health applications.

Mechanical and Control Systems Engineer | Sina Robotics and Medical Innovators Co. Dec 2022 – Sep 2023

Tehran, Iran

- Developed PLC-based control logic for a telesurgical laparoscopic robotic platform using Beckhoff TwinCAT and IEC 61131-3 Structured Text, with MATLAB/Simulink validation of inverse kinematics and initialization workflows.
- Built a Python/Kivy graphical user interface on Raspberry Pi for surgeon-facing robot control, system guidance, warnings, and improved human-robot interaction during testing and demonstrations.
- Tested and demonstrated the system with 6 experienced surgeons and supported medical-student workshops through operator training, troubleshooting, and usability feedback collection.

Laboratory Research Assistant | New Technology Research Center, Amirkabir University of Technology Nov 2020 – Nov 2022

Tehran, Iran

- Built experimental setups and prepared lab systems for robotics and mechatronics projects involving parallel manipulators, Delta robots, exoskeletons, CNC machines, 3D printers, CAD/MATLAB workflows, hardware setup, and technical documentation.
- Mentored 6 undergraduate students across 5 research teams, trained students on lab equipment and workflows, and supported technical organization for the 9th International Conference on Robotics and Mechatronics.

EDUCATION

Ph.D. in Mechanical Engineering | University of Maine, Orono, ME Expected Dec 2027

GPA: 4.00/4.00 | Dissertation: Smart Real-Time Gait Training System

M.S. in Mechanical Engineering | Amirkabir University of Technology (Tehran Polytechnic) Feb 2020

GPA: 3.65/4.00 | Thesis: Trajectory optimization of a robot arm in joint and Cartesian spaces

B.S. in Mechanical Engineering | University of Tehran Jul 2017

SELECTED HONORS

Flagship Fellowship Award, University of Maine Sep 2024 – Sep 2025

Graduate Student Government Award for conference proposal Mar 2025

PUBLICATIONS

Six peer-reviewed journal publications in wearable sensing, biomechanics, rehabilitation engineering, and robotics.

1. **Sharafian, E.**, & Hejrati, B. "Real-Time Foot Height Estimation and Activity Classification Using a Foot-Mounted IMU Implemented on a Smartphone." Sensors, 2026.
2. **Sharafian, E.**, Ellis, C., & Hejrati, B. "Real-Time Activity Recognition Using Minimal Biomechanical Features: A Lightweight IMU-Based Classifier for Older Adults." IEEE Access, 2025.
3. **Sharafian, E.**, et al. "The Effects of Real-Time Haptic Feedback on Gait and Cognitive Load in Older Adults." IEEE Transactions on Neural Systems and Rehabilitation Engineering, 2025.
4. Noghani, M. A., **Sharafian, E.**, Sidaway, B., & Hejrati, B. "Increasing Thigh Extension with Haptic Feedback Affects Leg Coordination in Young and Older Adult Walkers." Journal of Biomechanics, 2025.
5. **Sharafian, E.**, Ghassabzadeh, M., & Taghvaeipour, A. "Trajectory Optimization of a Spot-Welding Robot in the Joint and Cartesian Spaces." International Journal of Robotics and Automation, 2023.
6. **Sharafian, E.**, Taghvaeipour, A., & Ghassabzadeh, M. "Revisiting Screw Theory-Based Approaches in the Constraint Wrench Analysis of Robotic Systems." Robotica, 2022.